ScaDS.AI DRESDEN LEIPZIG

NFDI4 BIOIMAGE

# TRAINING: Data Science and AI for Medicine Training School 2026
## Day 2: RDM & DL Basics – Theory and Practice

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Gefördert durch:

Bundesministerium für Forschung, Technologie und Raumfahrt

SACHSEN

Diese Maßnahme wird gefördert durch die Bundesregierung aufgrund eines Beschlusses des Deutschen Bundestages. Diese Maßnahme wird mitfinanziert durch Steuermittel auf der Grundlage des von den Abgeordneten des Sächsischen Landtags beschlossenen Haushaltes.

Using materials from Robert Haase (DSC ScaDS.AI / Leipzig University)
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# AGENDA

1. Theory and Terms
   - FAIR-Principles

1. Medical Data

1. Practice
   - Data Types & Principles in RDM

1. MII Academy

Sneak Preview ;-)

# Basics

# Research Data Management (RDM)

**What does it really mean?**

# Research Data Management (RDM)

## IT & Cybersecurity

access control storage

data protection backup

## Legal Studies

copyright licensing

## Ethics

consent privacy

transparency

research integrity

## Data Science

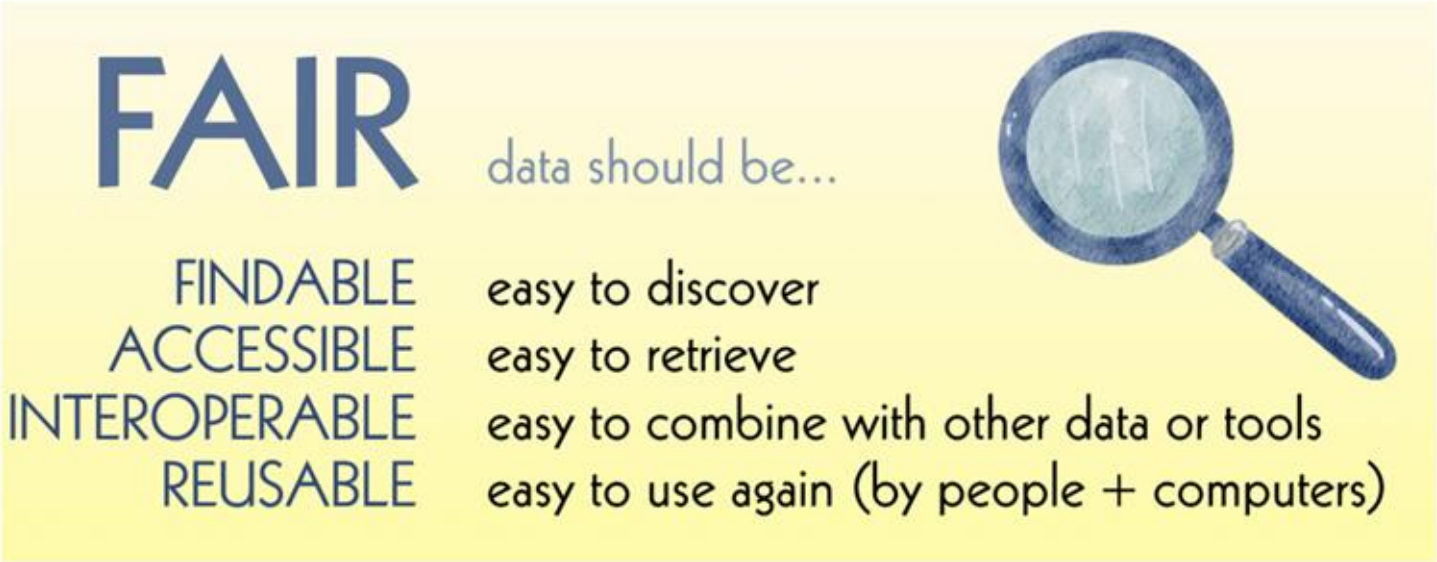
data quality curation

metadata analysis/ AI readiness

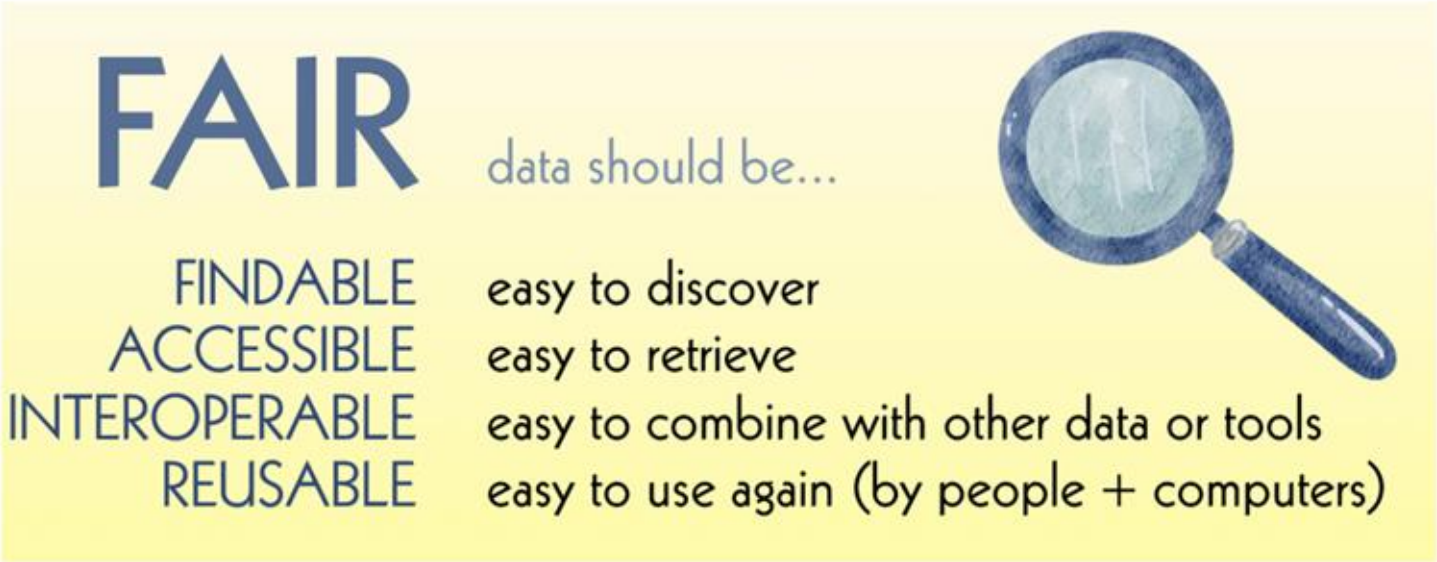
## Project Management

planning documentation workflows

# Principles in RDM



# Principles in RDM




As **open** as possible, as **closed** as necessary!

# Guidelines

1) Register datasets in public repositories
   - use trusted repositories with clearly defined access conditions and option to select a license
   - choose a repo that assigns a **DOI** (**= permanent** string **uniquely** identifying a digital object)

https://doi.org/10.1038/s41392-020-00243-2

DOI

URL

# ***Everything*** **is data but <u>where</u> can I upload it?**

# Data Repositories

GitHub

for code

figshare

for images

zenodo

for posters, presentations

Hugging Face

for DL models, datasets

COMING SOON

→ next talks

special considerations
for medical data

# Guidelines

2) Provide documentation & metadata (= data about data)
   - describe context, data structure, variables, methods using metadata schemas like Dublin Core
   - use tools like ReadMe files, Jupyter Notebooks

→ watch out for open standard, file formats (e.g. CSV, TXT, JSON, XML, PDF) & schemas!
→ watch out for consistent file naming & version control!

# All of this is possible...

| Name | Date modified | Type | Size |
|---|---|---|---|
| [illegible] | | File folder | |
| [illegible] | | File folder | |

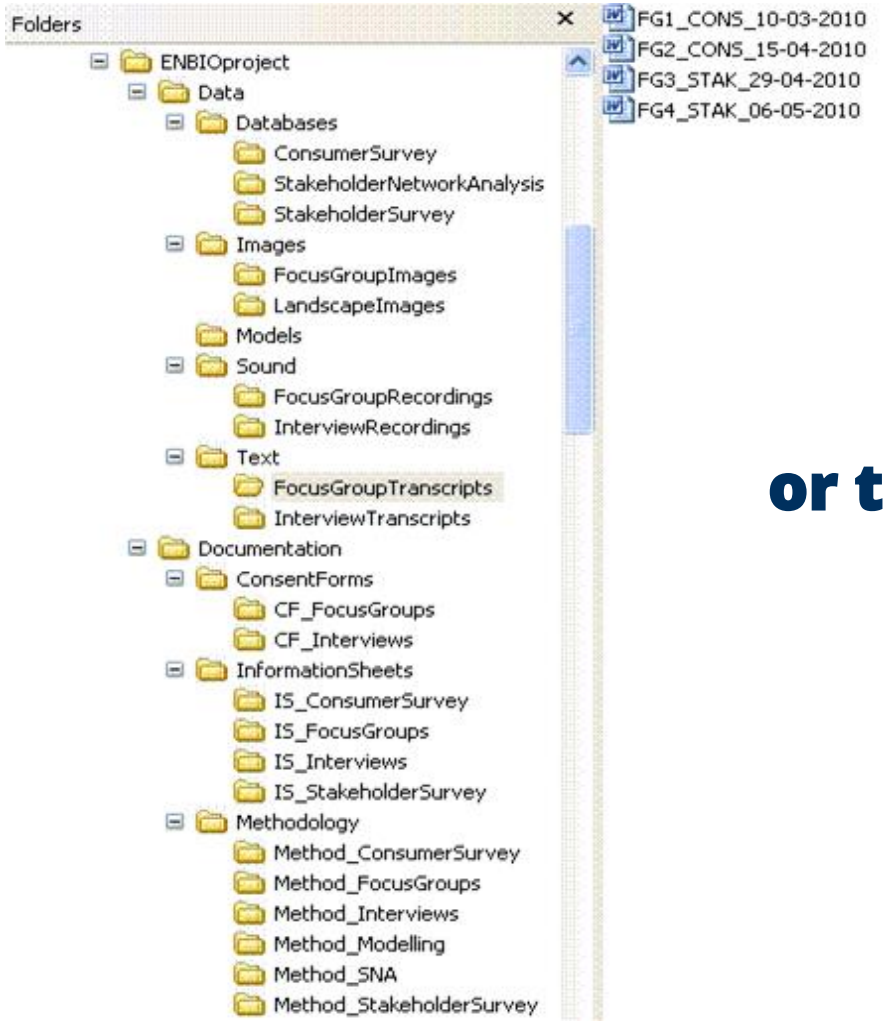
4B7C3646463D2D6B0139D159360A9D91_02_07-adverbs.jpg

6E839ED619188BAD4B4FF602819487B7_03_02-possessive.jpg

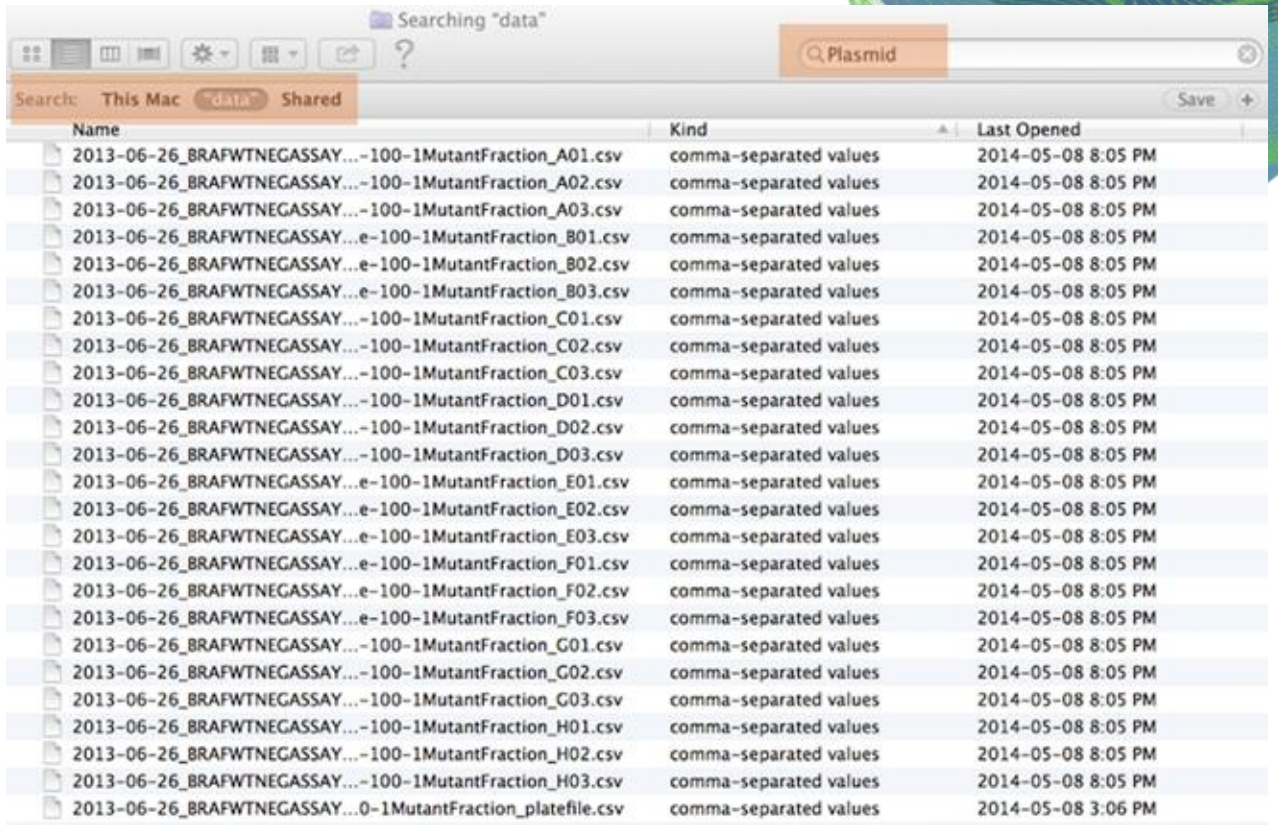
03_08-commas.mp4

03_09-minormarks.mp4

# … but please stick to this:




**or this**



# INCENTIVES!

**Increased visibility.**

Higher citation of data and publications.

More recognition in the field.

**Easier collaboration.**

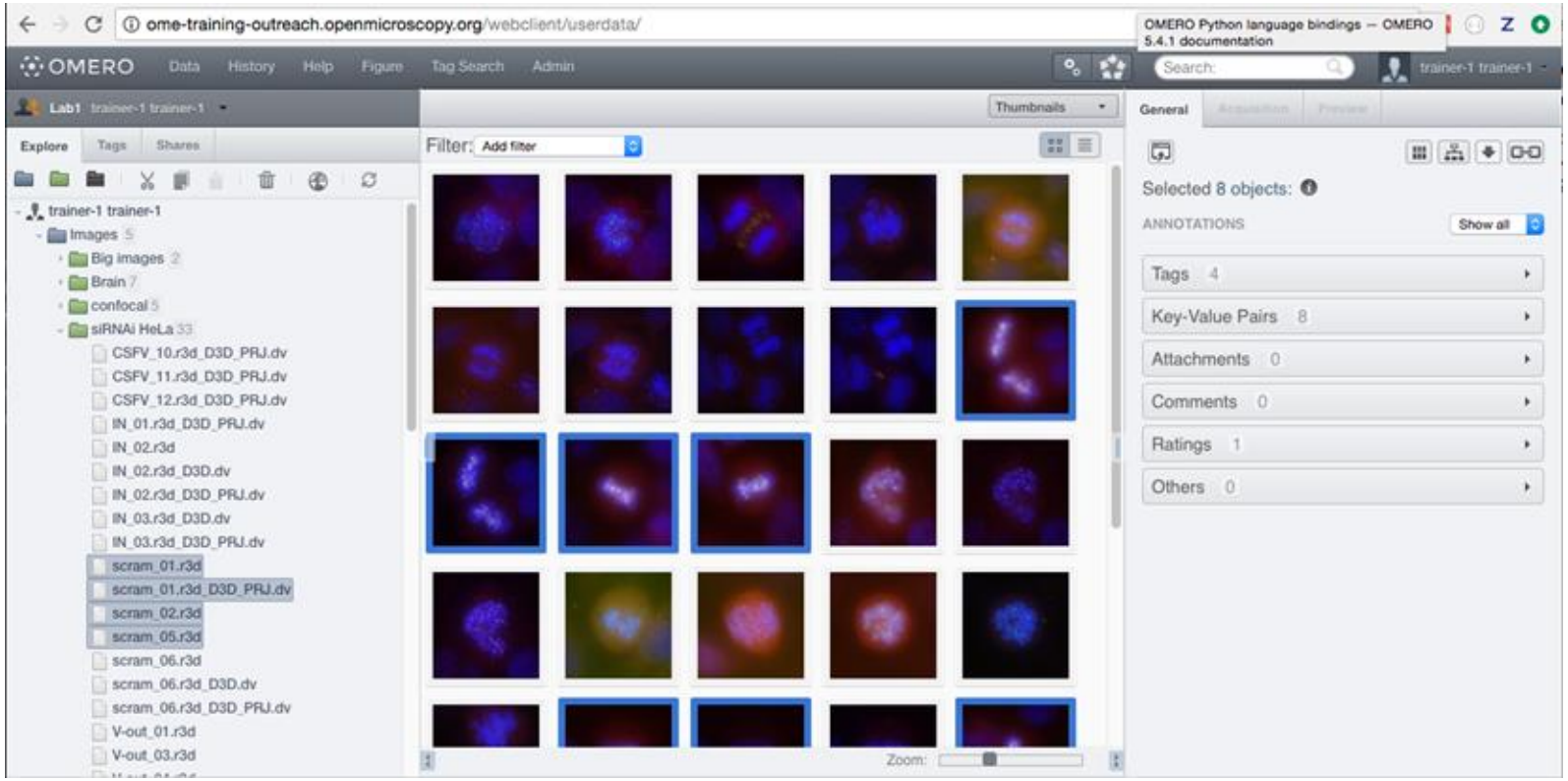
**Fostering new discoveries.**

**Saves time.**

# RDM TOOLS

# OMERO

= An **open-source** software for **image data** management




https://omero-guides.readthedocs.io/projects/introduction/en/latest/data-management.html

**Riccardo Massei**

https://zenodo.org/records/16880913

Tutorials & Help

# REDCap

= A **secure** web application for building and managing **survey data**

REDCap
Research Electronic Data Capture

| Institutions | Countries | Projects | Users | Citations |
|---|---|---|---|---|
| 8227 | 164 | 2.7M | 4.2M | 52.6k |

## Christof Meigen

Contact & Help

# Further Reading

**Guides**

- RDM-Guide (Uni Leipzig):
  https://www.uni-leipzig.de/fileadmin/ul/Grundsätze_Forschungsdatenmanagement.pdf

- Medical RDM-Guide (DFG):
  https://zenodo.org/records/4912388#.YPazF5dxfcu

**Contact Points**

- RDM-Services (Uni Leipzig)
  https://www.uni-leipzig.de/forschung/forschungsservice/forschungsdatenmanagement

- RDM-Services (Saxony)
  https://saxfdm.de/

- RDM-Services (Germany)
  https://www.nfdi.de/consortia/

# Acknowledgments

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